THE GOLDEN MOMENT AND CERTIFIED CONSTRAINTS ON AGRAWAL'S CONJECTURE AT $r = 5$

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Abstract. We prove, and formally verify in Lean, that the quadratic moment unit arising in Agrawal's congruence at $r = 5$ has quintic index

$$M_2 = 3\ \mathrm{ind}_5\left(\frac{1+\sqrt{5}}{2}\right).$$

Thus the quadratic moment is exactly the classical quintic character of the golden unit, up to the scalar 3. This note also collects the mechanically verified constraints obtained around that bridge. Every statement called a theorem below is proved in Lean 4 over Mathlib, with no `sorry`, no `admit` and no added axioms; the corresponding declaration name is given with each statement. Every computational claim carries a replayable certificate with its detection criterion embedded and SHA-256 manifests. We pose, without any claim of proof, the two open problems that remain central. Nothing in this note asserts Agrawal's conjecture.

1. Setting

Agrawal's conjecture (arising from the AKS primality test [1]; see also [2, 4, 5]) is stated for coprime positive integers n, r. We use its prime-r specialization: if r is prime, $r \nmid n$, and $(X-1)^n \equiv X^n - 1 \pmod{n,\, X^r - 1}$, then n is prime or $n^2 \equiv 1 \pmod{r}$. Our study concerns $r = 5$. This note isolates its mechanically certified core; the statements are elementary, and their interest lies in the role they play in the wider study and in the fact that a proof assistant has checked every one of them.

Local notation and all parity hypotheses are introduced at the point of use. In the first sections, let

$$R_p \;=\; (\mathbf{Z}/p)[X]\,/\,(X^2 - X - 1), \qquad \varepsilon \;=\; \text{the class of } X,$$

so that $\varepsilon^2 = \varepsilon + 1$, and set $\sqrt{5} := 2\varepsilon - 1$, which satisfies $(\sqrt{5})^2 = 5$ in R_p (`GoldenRing`, `eps_sq`, `sqrt5_sq`). Let F_n, L_n denote the Fibonacci and

This is a human-directed, LLM-assisted project. The mathematical exploration and formalization were produced by an orchestrated multi-model pipeline (Claude and GPT-family models), directed by Daniele Cappello and subjected to independent model audits. Every formal claim is checked by the Lean kernel; model agreement is not treated as evidence. Repository with the Lean sources and all computational certificates: `https://github.com/Solarys431/agrawal-r5`.

Lucas numbers, with $L_n = F_{n+1} + F_{n-1}$ (`lucas`). Define

$$A_n = \begin{cases} L_n, & n \text{ even}, \\ F_n, & n \text{ odd}, \end{cases} \qquad H_n = \gcd\bigl(A_n,\, 5^{\,n-1}+1\bigr),$$

and, dually, $B_n = F_n$ (n even), L_n (n odd), with $J_n = \gcd\bigl(B_n,\, 5^{\,n-1}-1\bigr)$.

2. The golden Frobenius and the inertia of J_n

Theorem 2.1 (`golden_frobenius`). *Assume* 5 *is not a square in* $\mathbf{Z}/p$ *and* $p \neq 2$. *Then* $\varepsilon^p = 1 - \varepsilon$ *in* R_p.

Certified proof sketch. By Euler's criterion (`euler_nonsquare`) we have $5^{(p-1)/2} = -1$ in $\mathbf{Z}/p$, hence

$$(\sqrt{5})^p = (5)^{(p-1)/2}\sqrt{5} = -\sqrt{5} \qquad (\texttt{sqrt5\_pow\_p}).$$

Expanding $(2\varepsilon-1)^p$ by the Frobenius endomorphism and cancelling the unit 2 gives the claim. □

Corollary 2.2 (`golden_pow_p_succ`). *Under the same hypotheses,* $\varepsilon^{\,p+1} = -1$ *in* R_p.

Theorem 2.3 (`inertia_J`). *Assume* 5 *is not a square in* $\mathbf{Z}/p$ *and* $p \neq 2$. *Then for no* $n \geq 1$ *do* $\varepsilon^{2n} = 1$ *in* R_p *and* $5^{\,n-1} = 1$ *in* $\mathbf{Z}/p$ *hold simultaneously.*

Certified proof sketch. If n is odd, raising $\varepsilon^{2n} = 1$ to the power $(p+1)/2$ and comparing with $(\varepsilon^{p+1})^n = (-1)^n = -1$ gives $1 = -1$, absurd for $p \neq 2$. If n is even, $5 = 5^{\,n} = (5^{\,n/2})^2$ exhibits 5 as a square, against the hypothesis. □

Via the identity $\varepsilon^{\,n+1} = F_{n+1}\varepsilon + F_n$ in R_p (`golden_pow_fib` and its predecessor form `golden_pow_pred`) one obtains the divisibility forms, in which the hypothesis is purely arithmetic by quadratic reciprocity (`not_isSquare_five`):

Theorem 2.4 (`inertia_J_fib_mod5`, `inertia_J_lucas_mod5`). *Let* $p \equiv \pm 2 \pmod 5$, $p \neq 2$. *For even* $n \geq 2$, p *does not divide* $\gcd(F_n,\, 5^{\,n-1}-1)$*; for odd* $n \geq 1$, p *does not divide* $\gcd(L_n,\, 5^{\,n-1}-1)$. *Equivalently, no* odd *prime inert in* $\mathbf{Q}(\sqrt{5})$ *divides any* J_n.

3. The canonical support witness

Theorem 3.1 (`support_witness`). *Let* $p \equiv 2 \pmod 5$, $p \neq 2$, *and put* $n_0 = (p+1)/2$. *Then* $n_0 \equiv 4 \pmod 5$, $p \mid 5^{\,n_0-1}+1$, *and* p *divides* L_{n_0} *if* n_0 *is even,* F_{n_0} *if* n_0 *is odd. In particular* $p \mid H_{n_0}$, *with an index in the class* 4 mod 5.

4. The moment obstruction

Let F be a finite field, let $e : F^\times \to F$, and define

$$M_j(e) = \sum_{a \in F^\times} a^j e(a).$$

The discrete logarithms occurring in the local Agrawal problem satisfy a covariance relation of the following form.

Theorem 4.1 (`moment_covariance`)**.** *If $t \in F^\times$ and*

$$e(ta) = t\, e(a) \qquad (a \in F^\times),$$

then

$$tM_j(e) = t^{-j} M_j(e).$$

Certified proof sketch. Multiply the defining sum by t, use the covariance relation, and substitute $b = ta$. Multiplication by t permutes $F^\times$, so

$$tM_j = \sum_a a^j e(ta) = \sum_b (t^{-1}b)^j e(b) = t^{-j} M_j.$$

The change of variables is implemented by an explicit finite equivalence in the Lean proof. $\square$

Corollary 4.2 (`pow_succ_eq_one_of_moment_ne_zero`)**.** *Under the same covariance relation,*

$$M_j(e) \neq 0 \quad \Longrightarrow \quad t^{j+1} = 1.$$

Without the nonvanishing hypothesis one has the factor identity

$$\bigl(t^{j+1} - 1\bigr) M_j(e) = 0 \qquad (\textit{moment\_obstruction}).$$

This is the kernel-checked Fourier argument behind the moment obstruction $\mathrm{im}_r(S) \subseteq T$ in the wider local theory. The present file formalizes its algebraic engine; connecting the abstract function e to the complete definition of $S(p, r)$ remains outside this core.

5. The golden moment

Let ζ satisfy $\Phi_5(\zeta) = 0$, and put

$$s = 1 + 2(\zeta + \zeta^4), \qquad \varepsilon = 1 + \zeta + \zeta^4, \qquad U_2 = \prod_{a=1}^{4} (\zeta^a - 1)^{a^2 \bmod 5}.$$

Thus $s^2 = 5$, $\varepsilon = (1 + s)/2$ whenever 2 is invertible, and

$$U_2 = (\zeta - 1)(\zeta^2 - 1)^4(\zeta^3 - 1)^4(\zeta^4 - 1).$$

The quintic index of U_2 is the quadratic moment M_2 in the moment decomposition of the local Agrawal congruence.

Theorem 5.1 (`golden_moment_factorization`). *In every commutative ring in which $\zeta^4+\zeta^3+\zeta^2+\zeta+1=0$, one has*

$$\boxed{U_2 = s^5\varepsilon^3.}$$

Certified proof sketch. Set

$$A = (\zeta-1)(\zeta^4-1), \qquad B = (\zeta^2-1)(\zeta^3-1).$$

Direct reductions modulo Φ_5, formalized without division, give

$$AB = 5, \qquad B = s\varepsilon, \qquad s^2 = 5.$$

Since $U_2 = AB^4 = (AB)B^3$, it follows that $U_2 = 5(s\varepsilon)^3 = s^5\varepsilon^3$. □

Theorem 5.2 (Golden index theorem). *Let p be prime with $5 \mid p-1$, let $\zeta \in \mathbf{F}_p$ satisfy $\Phi_5(\zeta)=0$, and suppose that the displayed elements are units. For the discrete quintic index $\mathrm{ind}_5 : \mathbf{F}_p^\times \to \mathbf{Z}/5\mathbf{Z}$ constructed in* `quinticIndex`*,*

$$\boxed{M_2 = 3\ \mathrm{ind}_5(\varepsilon).}$$

The same identity holds for every multiplicative quintic character.

Lean declarations: `golden_moment_index`, `zmod_golden_moment_index`.

Certified proof sketch. Every quintic index kills the fifth power s^5, and sends ε^3 to $3\ \mathrm{ind}_5(\varepsilon)$. Apply Theorem 5.1. □

Remark 5.3 (Prior art and novelty boundary). Williams and Hardy computed the quintic index of the golden unit in Dickson coordinates [6, Theorem 5]; that classical character is not claimed here as new. The contribution of Theorems 5.1–5.2 is the identification of the *Agrawal quadratic moment* with that character. We have not found this bridge in the targeted literature search, but priority remains provisional pending specialist review.

6. The golden–cyclotomic entanglement

Theorem 6.1 (`entanglement`). *In any commutative ring, if $\zeta^4+\zeta^3+\zeta^2+\zeta+1=0$, then*

$$\zeta^3\,(1+\zeta+\zeta^4)^2\,(\zeta-1)^4 \;=\; 5.$$

In particular this holds in $\mathbf{Z}[\zeta_5]$ with $\varepsilon = 1+\zeta+\zeta^4$ the image of the golden unit.

The proof is a single `linear_combination` with an explicit cofactor; $\zeta^5 = 1$ follows from the same hypothesis (`zeta_pow_five`).

7. The Fermat-shadow glue

Lemma 7.1 (`mul_dvd_gcd_mul`). *If $x \mid M$ and $y \mid M$, then $xy \mid \gcd(x,y)\,M$.*

Theorem 7.2 (`fermat_shadow`)**.** *Let n be squarefree with $5 \nmid n$, and suppose that for every prime $q \mid n$*

$$(*) \qquad 5^{n/q-1} \equiv 1 \pmod{q}.$$

Then $n \mid 5^{n-1} - 1$. Thus n satisfies the base-5 Fermat congruence; if n is composite, it is a Fermat pseudoprime to base 5.

Remark 7.3. Hypothesis $(*)$ is the local norm constraint of the wider study; in Theorem 7.2 it is an explicit hypothesis. The next theorem removes it: in our squarefree $r = 5$ setting, the Agrawal congruence itself forces the base-5 Fermat congruence.

Theorem 7.4 (`agrawal_fermat_shadow`)**.** *Let n be squarefree with $5 \nmid n$, and suppose the Agrawal congruence at $r = 5$ holds: $(X-1)^n \equiv X^n - 1 \pmod{n, X^5 - 1}$. Then $n \mid 5^{n-1} - 1$. In particular, every squarefree composite counterexample to Agrawal's conjecture at $r = 5$ prime to 10 is a Fermat pseudoprime to base 5.*

Certified proof sketch. Reduce the congruence mod a prime $q \mid n$; it becomes a polynomial identity in $(\mathbf{Z}/q)[X]$, which can be evaluated at the four elements z^j $(j = 1, \dots, 4)$ of the quotient ring $(\mathbf{Z}/q)[X]/\Phi_5$ (not a field in general), where $z^5 = 1$ kills the modulus. Multiplying the four evaluated identities and using the explicit-cofactor identity $(z-1)(z^2-1)(z^3-1)(z^4-1) = 5$ (`prod_pow_sub_one`) gives $5^n = 5$ in the quotient ring; injectivity of the constant embedding descends this equality to $\mathbf{Z}/q$, hence $5^{n-1} = 1$ there (`agrawal_local`), and the squarefree gluing finishes. No Frobenius descent and no norm maps are needed. □

Remark 7.5 (Prior art). Lenstra observed the more general implication $r^n \equiv r \pmod{n}$ directly from Agrawal's congruence [3, Remark 5]. Theorem 7.4 is therefore not claimed as new mathematics: its contribution here is an end-to-end Lean formalization of the squarefree $r = 5$ specialization.

8. The two-adic jaw

The bridge constrains a squarefree solution of the Agrawal congruence only through the multiplicative order of 5. On the primes that are inert in $\mathbf{Q}(\sqrt{5})$ that constraint is rigid at the prime 2.

Theorem 8.1 (`agrawal_two_adic_jaw`)**.** *Let n be squarefree with $5 \nmid n$ and suppose the Agrawal congruence at $r = 5$ holds. Let $q \mid n$ be an odd prime with $q \equiv \pm 2 \pmod{5}$. Then*

$$v_2(q-1) \;\le\; v_2(n-1).$$

Certified proof sketch. By Theorem 7.4, $\operatorname{ord}_q(5)$ divides $n-1$. Since $q \equiv \pm 2 \pmod{5}$, quadratic reciprocity gives $(5/q) = -1$ (`not_isSquare_five`), so $5^{(q-1)/2} = -1$ (`euler_nonsquare`) and the order does *not* divide $(q-1)/2$. A divisor of $2^a u$ with u odd that fails to divide $2^{a-1}u$ is 2-adically saturated (`pow_two_dvd_of_not_dvd_half`), whence $v_2(\mathrm{ord}_q 5) = v_2(q-1)$, and the claim follows; the local form is `two_adic_jaw_local`. $\square$

Corollary 8.2 (`agrawal_inert_mod_four`)**.** *Under the same hypotheses, if* $n \equiv 3 \pmod 4$ *then every prime factor of* n *that is inert in* $\mathbf{Q}(\sqrt{5})$ *satisfies* $q \equiv 3 \pmod 4$.

Remark 8.3. The split primes behave in the opposite way: for $q \equiv \pm 1 \pmod 5$ one has $(5/q) = +1$, hence $\mathrm{ord}_q(5) \mid (q-1)/2$ and $v_2(\mathrm{ord}_q 5) < v_2(q-1)$, so no constraint is transmitted. Both statements above are unconditional.

9. The bounded order

The bridge and the jaw both read the congruence through the multiplicative order of an element. The element that carries the congruence itself is $\zeta - 1$, where ζ is the class of X in $T = \mathbf{F}_p[X]/\Phi_5$. A priori its order is only known to divide $|T^\times| = p^4 - 1$. It divides much less.

Theorem 9.1 (`order_bounded`)**.** *Let* p *be a prime with* $p \equiv \pm 2 \pmod 5$*, so that* Φ_5 *stays irreducible modulo* p*, and let* ζ *be the class of* X *in* $T = \mathbf{F}_p[X]/\Phi_5$*. Then*

$$(\zeta - 1)^{10p^2} = (\zeta - 1)^{10}.$$

The displayed identity is the statement checked by the cited Lean declaration. Mathematically, irreducibility makes T a field and $\zeta - 1 \neq 0$; cancellation therefore gives the finite-field corollary that the multiplicative order of $\zeta - 1$ divides $10(p^2 - 1)$. This order formulation is not bundled into `order_bounded`.

Certified proof sketch. Write $u = \zeta - 1$. In characteristic p the Frobenius is additive, so $u^p = \zeta^p - 1$, and since $\zeta^5 = 1$ the exponent only matters modulo 5 (`zeta5_pow_mod`), giving $u^p = \zeta^{p \bmod 5} - 1$ (`zeta_sub_one_pow_p`). Applying the Frobenius once more multiplies that exponent by p again, and here the two inert classes merge: $2^2 \equiv 3^2 \equiv 4 \pmod 5$, so in both cases $u^{p^2} = \zeta^4 - 1$ (`zeta_sub_one_pow_sq`). Finally $\zeta^4 = \zeta^{-1}$ turns that into

$$u^{p^2} = \zeta^{-1} - 1 = -(\zeta - 1)\zeta^{-1} = -\zeta^4\, u$$

(`order_bound_relation`): the square of the Frobenius merely scales u by $-\zeta^4$. Raising to the tenth power kills the scalar, since $(-1)^{10} = 1$ and $\zeta^{40} = (\zeta^5)^8 = 1$. $\square$

Remark 9.2. The statement is phrased as $u^{10p^2} = u^{10}$ rather than as a divisibility of the order, so that no invertibility of u is needed. It is the form actually used downstream: the factor 10 is what forces the modulus 80 to appear in the congruence conditions on the prime factors of a putative counterexample, and it is what makes an exact-order computation factor a ten-digit integer instead of a twenty-digit one.

10. An elementary box

The following identity is used to confine the largest prime factor of a three-factor solution to an interval. It is stated over any commutative ring and proved by expansion.

Theorem 10.1 (Box identity)**.** *In any commutative ring, for all p, q, r,*

$$2\big[(pq-1)(pr-1)(qr-1)-(p^2-1)(q^2-1)(r^2-1)\big]$$
$$=(p^2-1)(q-r)^2+(q^2-1)(p-r)^2+(r^2-1)(p-q)^2.$$

In particular the left-hand side is non-negative whenever $p, q, r \geq 1$.
Lean declarations:
`prod_pairs_sub_prod_squares`,
`prod_pairs_ge_prod_squares`.

Combined with `lt_two_mul_of_sq_le`, which turns $r^2 \leq 2I(pq)$ with $q < r$ into $r < 2Ip$, this bounds the third prime by an explicit multiple of the smallest one.

11. The Lenstra–Pomerance proposition

The statement below is the one on which the search for counterexamples at $r = 5$ rests. It appears in the AIM notes of Lenstra and Pomerance [2, pp. 30–32] and is quoted throughout the subsequent literature; to our knowledge it had not been machine-checked before.

Mathlib does not currently contain Carmichael numbers or Korselt's criterion, so those had to be introduced: `IsCarmichael` and `IsLucasCarmichael` package the two divisibility conditions used below. We also prove that a Carmichael number is a Fermat pseudoprime to every coprime base (declaration `IsCarmichael.fermatPsp`). This implication is *not* used in Theorem 11.1, which needs only the divisibility conditions themselves.

Theorem 11.1 (`lenstra_proposition_card`)**.** *Let n be squarefree, suppose the number k of its prime factors satisfies $k \equiv 1 \pmod 4$, suppose every prime $p \mid n$ satisfies $p \equiv 3 \pmod{80}$, and suppose Korselt's two conditions hold: $(p-1) \mid (n-1)$ and $(p+1) \mid (n+1)$ for every $p \mid n$. Then*

$$(X-1)^n \equiv X^n - 1 \pmod{n,\ X^5-1} \qquad \textit{and} \qquad n^2 \not\equiv 1 \pmod 5.$$

These are the hypotheses of the original statement. That $n \equiv 3 \pmod{80}$ follows from them is itself proved (`mod_eighty_of_card`): for squarefree n the product of the prime factors is n, each factor is 3 modulo 80, and $3^4 = 81 \equiv 1 \pmod{80}$ reduces the exponent to the class of k modulo 4. A variant assuming $n \equiv 3 \pmod{80}$ directly is available as `lenstra_proposition`.

The second conclusion is what makes such an n, *if composite*, a genuine counterexample: Agrawal's conjecture allows the escape $n^2 \equiv 1 \pmod r$, and here it is closed (`sq_not_one_mod_five`). The proviso matters: primes satisfy the hypotheses trivially—$n = 83$ is one—and are of course not counterexamples. Only a composite witness would be one, and none is known.

Certified proof sketch. The route is the one of the original proof, which already contains the identity $(\zeta_5 - 1)^{p^2} = -\zeta_5^{-1}(\zeta_5 - 1)$, the resulting bound on the order, and the reduction to $n \equiv p$; we repackage it through a single observation: *the three conditions imposed on n are satisfied by p itself*, since $p \equiv 1 \pmod{p-1}$, $p \equiv -1 \pmod{p+1}$ and $p \equiv 3 \pmod{80}$. Hence

$$n \equiv p \pmod{\operatorname{lcm}(p-1,\, p+1,\, 80)}, \qquad \operatorname{lcm}(p-1,\, p+1,\, 80) = 10(p^2-1)$$

(`lcm_three_eq`, `sub_dvd_of_korselt`), and that modulus is exactly the one bounding the order of $\zeta - 1$ by Theorem 9.1. Therefore $(\zeta-1)^n = (\zeta-1)^p$, and the Frobenius gives $(\zeta-1)^p = \zeta^p - 1 = \zeta^3 - 1 = \zeta^n - 1$, which is the congruence in the cyclotomic component (`lenstra_local`). On the component $X \mapsto 1$ both sides vanish, and since $X^5 - 1 = (X-1)\Phi_5$ with the two factors coprime for $p \neq 5$, the two pieces glue to the congruence modulo p (`agrawal_mod_p`, using `dvd_of_dvd_both`). Finally, for squarefree n a congruence holding modulo every prime factor holds modulo n: dividing by the monic $X^5 - 1$ over $\mathbf{Z}$, every coefficient of the remainder is divisible by each $p \mid n$, hence by n (`congruence_of_local`). ☐

Remark 11.2. Nothing here asserts that a composite such n exists. A composite number satisfying the hypotheses would be simultaneously a Carmichael and a Lucas–Carmichael number; no such integer is known. Williams posed this existence problem in 1977 [8, 10]; Pomerance's 1984 Baillie–PSW heuristic later imposed the same paired Korselt conditions [7]. Note also that the proposition gives a *sufficient* condition for producing a counterexample, not a necessary one: it does not say where all counterexamples must lie. It certifies the tool used to look for them; it does not decide Agrawal's conjecture in either direction.

12. The forced partition

The simultaneous Carmichael and Lucas–Carmichael conditions impose a prime-by-prime separation on the factors of any hypothetical witness.

Theorem 12.1 (`common_divisor_forced`)**.** *Let $n = pm = p'm'$, and suppose that both Korselt divisibilities hold at p and at p':*

$$p-1 \mid n-1, \quad p+1 \mid n+1, \qquad p'-1 \mid n-1, \quad p'+1 \mid n+1.$$

If $d \mid p-1$ and $d \mid p'+1$, then $d \mid 2$. Consequently, for an odd prime q, the two sets

$$\{p \mid n : q \mid p-1\}, \qquad \{p \mid n : q \mid p+1\}$$

cannot both be nonempty (`partition_forced`).

Certified proof sketch. Writing $n = pm$, the two divisibilities at p give $p-1 \mid m-1$ and $p+1 \mid m-1$ (`dvd_sub_of_korselt_pair`). Since $d \mid p-1$, it follows that $d \mid m-1$, and therefore $n \equiv p \pmod{d}$. The same argument at p', using $d \mid p'+1$, gives $n \equiv p' \pmod{d}$, so $p \equiv p' \pmod{d}$. But the two side conditions give $p \equiv 1 \pmod{d}$ and $p' \equiv -1 \pmod{d}$. Thus $d \mid 2$. ☐

Remark 12.2 (Scope and prior art). The compatibility condition $\gcd(p_i - 1, p_j + 1) = 2$ for Williams sets is stated explicitly by McIntosh [9, §4]; it is not a new mathematical claim of this paper. The contribution here is a kernel-checked arbitrary-divisor form and its reusable corollaries. The intermediate congruence $n \equiv p \pmod{(p^2-1)/2}$, and the stronger consequence that all prime factors of a simultaneous Carmichael and Lucas–Carmichael number are congruent modulo 12, already appear in Leng's 2017 study [11]; the named mod-3 corollary is `three_congruence_forced`. Theorem 12.1 is a useful structural sieve, not a proof that simultaneous numbers do not exist. In particular, the exclusions contributed by different q need not be statistically independent.

There is also no contradiction from size alone. If $L = \mathrm{lcm}_{p|n}(p-1)$ and $M = \mathrm{lcm}_{p|n}(p+1)$, then $LM \mid n^2 - 1$, but

$$LM \le \prod_{p|n}(p^2-1) < n^2$$

is automatic. The non-tautological information is instead at each prime-power level: if $I^-_{\ell,e} = \{i : \ell^e \mid p_i - 1\}$, then $\prod_{i \notin I^-_{\ell,e}} p_i \equiv 1 \pmod{\ell^e}$; if $I^+_{\ell,e} = \{i : \ell^e \mid p_i + 1\}$, then $\prod_{i \notin I^+_{\ell,e}} p_i \equiv (-1)^{|I^+_{\ell,e}|+1} \pmod{\ell^e}$. These incidence congruences, rather than the global inequality, are the next structural sieve.

The cubic parametrization used in the five-factor search does not provide an independent constraint: the strong complementary row $10(p^2-1) \mid m-1$ is equivalent, for $p \neq 0$, to

$$pm = p + 10u(p^3 - p)$$

for some u. Both directions are kernel-checked:

`cubic_signature_of_strong_row`
`strong_row_of_cubic_signature`.

13. Computational certificates

Independently of the kernel-checked statements above, the repository carries replayable certificates, each embedding its own detection criterion and reconstruction data, with SHA-256 manifests:

(1) seven *certified-empty fibres* of a three-factor sieve associated with the study (pairs $(13, 37)$, $(17, 113)$, $(13, 277)$, $(23, 167)$, $(7, 823)$, $(83, 347)$, $(463, 1327)$): for each pair, the finitely many levels required by the certificate are integrally factored, all prime factors are proven prime, and no prime in the certificate's detector classes survives;

(2) a self-certifying census of H_n for all $n \equiv 4 \pmod 5$, $n \le 10^5$: 9,725 nontrivial values, all factorizations proven, and *no* prime factor $\equiv \pm 1 \pmod 5$ occurs.

These are certificates of finite computations; they prove exactly what they state and nothing beyond, and both replay from the repository alone. Separately, the repository freezes as *hash-only provenance* (hashes, not artifacts) a SHA-256 index of the working material of the study, including a prime-by-prime corpus up to 10^9; nothing in this note relies on it.

14. The kernel form of the golden-inertia wall

The arithmetic reduction underlying the first open problem is now also kernel-checked. Recall the definitions of A_n and H_n from Section 1.

Theorem 14.1 (kernel-checked scalar profile). *Let p be prime, $p \neq 5$, and $n \geq 1$. Then*

$$p \mid H_n \quad \Longleftrightarrow \quad \varepsilon^{2n} = -1 \textit{ in } R_p \quad \textit{and} \quad 5^{\,n-1} = -1 \textit{ in } \mathbf{F}_p.$$

The declaration is

`dvd_goldenH_iff_scalar_profile`.

Both parity branches are proved in both directions. In particular, the Fibonacci–Lucas condition is not used merely as a necessary shadow of the scalar semiperiod.

Theorem 14.2 (kernel-checked 2-adic form). *Let $p \neq 2, 5$ be prime, $n \geq 2$, and $p \mid H_n$. Then*

$$v_2(\mathrm{ord}_p(5)) = v_2(n-1) + 1,$$

and

$$\left(\frac{5}{p}\right) = -1 \quad \Longleftrightarrow \quad v_2(p-1) = v_2(n-1) + 1.$$

The corresponding declarations are

`dvd_goldenH_order_factorization_two`
`dvd_goldenH_nonsquare_iff_two_adic_saturation`.

Thus the remaining universal assertion is exactly the last equality, not the already-proved order identity. The theorem is a reduction of the obstruction; it neither proves nor assumes golden inertia.

15. Two open problems

We pose the two statements that our working material identifies as the central open problems of the $r = 5$ study. We claim no proof of either statement.

Conjecture 15.1 (golden inertia). *For every $n \equiv 4 \pmod 5$, every odd prime factor of H_n distinct from 5 is $\equiv \pm 2 \pmod 5$.*

Theorem 3.1 shows that every prime $p \equiv 2 \pmod 5$ does occur in some H_n with $n \equiv 4 \pmod 5$; the census in item (2) above verifies Conjecture 15.1 for all $n \leq 10^5$. Theorem 14.2 gives its exact kernel-checked saturation form.

Conjecture 15.2 (fibre emptiness). *The pattern of item (1) above is general: for every admissible pair, the corresponding fibre is empty.*

16. Formalization data

At the release state audited for this version there are twenty-six modules and 3027 source lines; no `sorry`, no `admit`, and no axioms beyond Mathlib's. Mathlib is pinned to a fixed upstream commit; a clean clone builds with `lake exe cache get && lake build -wfail`. The table mapping each statement of this note to its declaration and file is in the repository `README`.

17. Reproducibility map

The repository is organized so that the three kinds of evidence never collapse into one another:

(1) the kernel layer is rebuilt with `lake build -wfail`; its flagship modules cover the moment obstruction, the golden moment, and the forced partition;
(2) the finite-computation layer is replayed by running `tools/verify_certificates.py`; the optional `-full` flag recomputes the complete H_n census;
(3) the literature layer is recorded claim by claim in `NOVELTY_AND_PRIOR_ART.md`, including negative novelty findings.

A read-only GitHub Actions workflow runs the first two checks with Lean warnings promoted to errors. The release gate remains private until a human specialist has reviewed the golden-moment priority claim.

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